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Recap on 2D kinematics

Kinematic Equations with Constant Acceleration

x-component

$$a_x = \text{constant}$$

$$v_x = v_{o_x} + a_x t$$

$$x = x_o + v_{o_x} t + \frac{1}{2} a_x t^2$$

y-component

$$a_y = \text{constant}$$

$$v_y = v_{o_y} + a_y t$$

$$y = y_o + v_{o_y} t + \frac{1}{2} a_y t^2$$

z-component

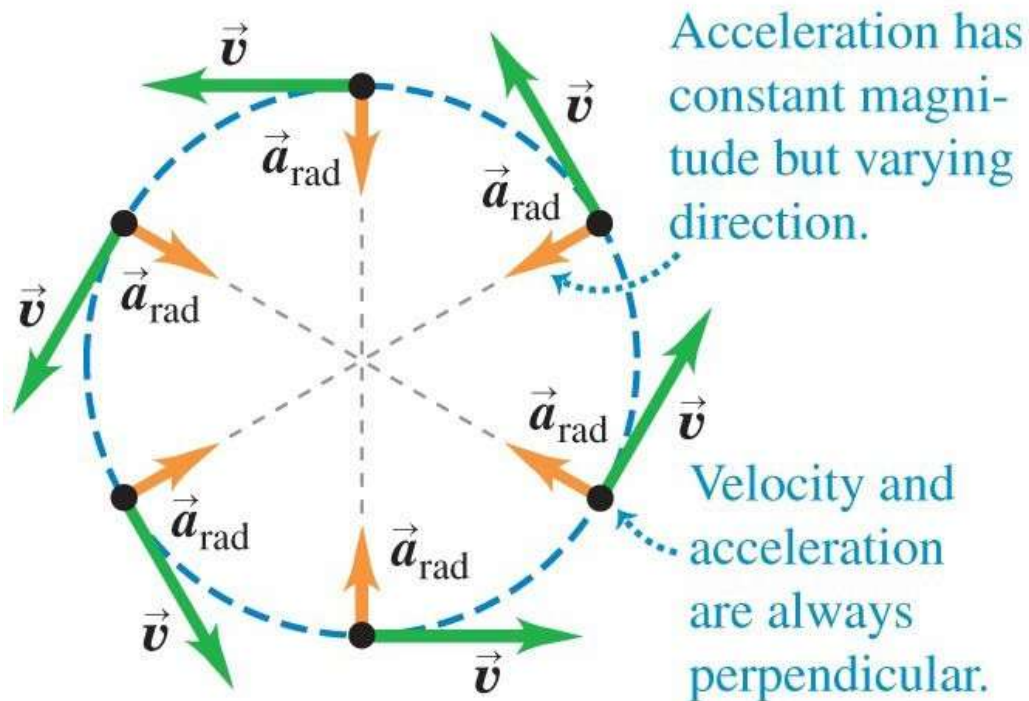
$$a_z = \text{constant}$$

$$v_z = v_{o_z} + a_z t$$

$$z = z_o + v_{o_z} t + \frac{1}{2} a_z t^2$$

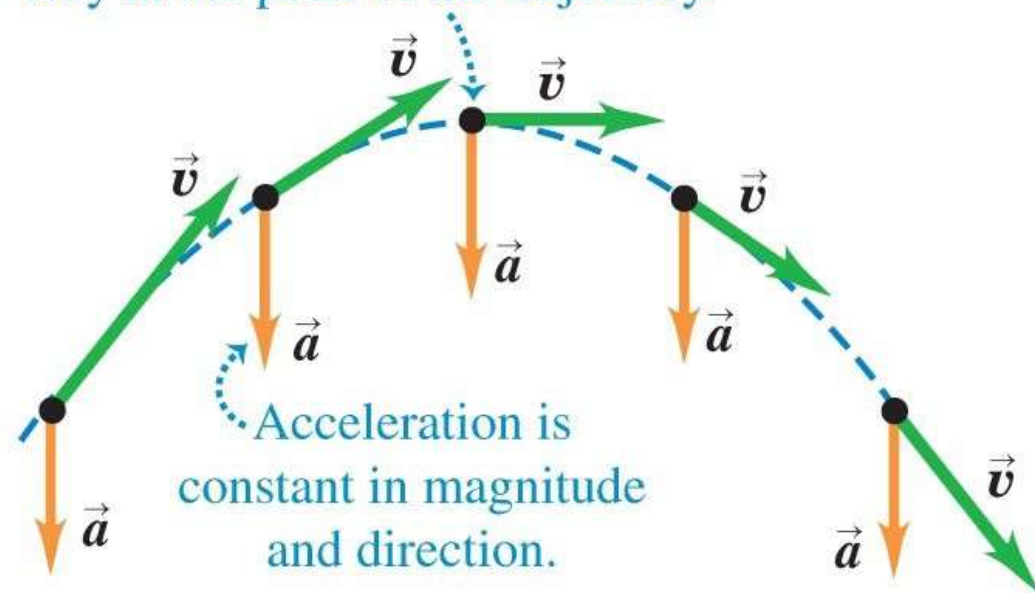
Two trajectories with constant acceleration

(a) Uniform circular motion



(b) Projectile motion

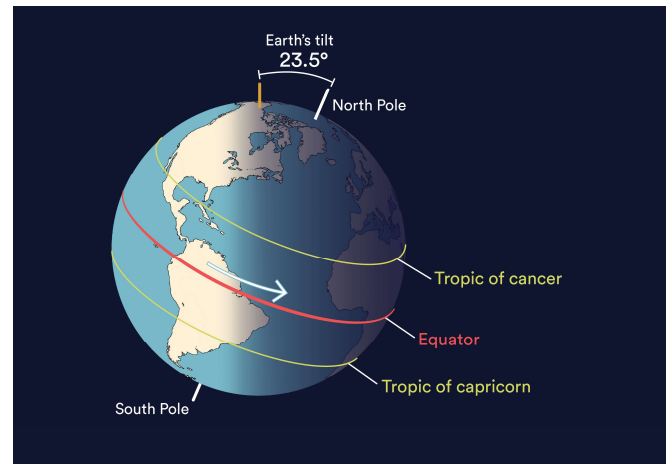
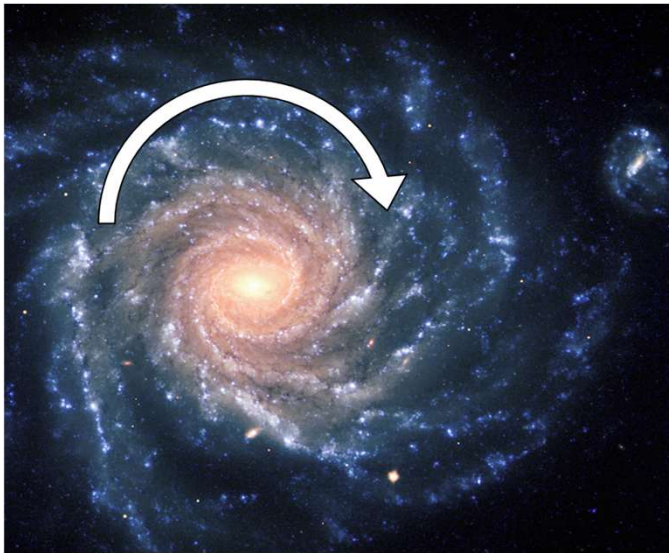
Velocity and acceleration are perpendicular only at the peak of the trajectory.





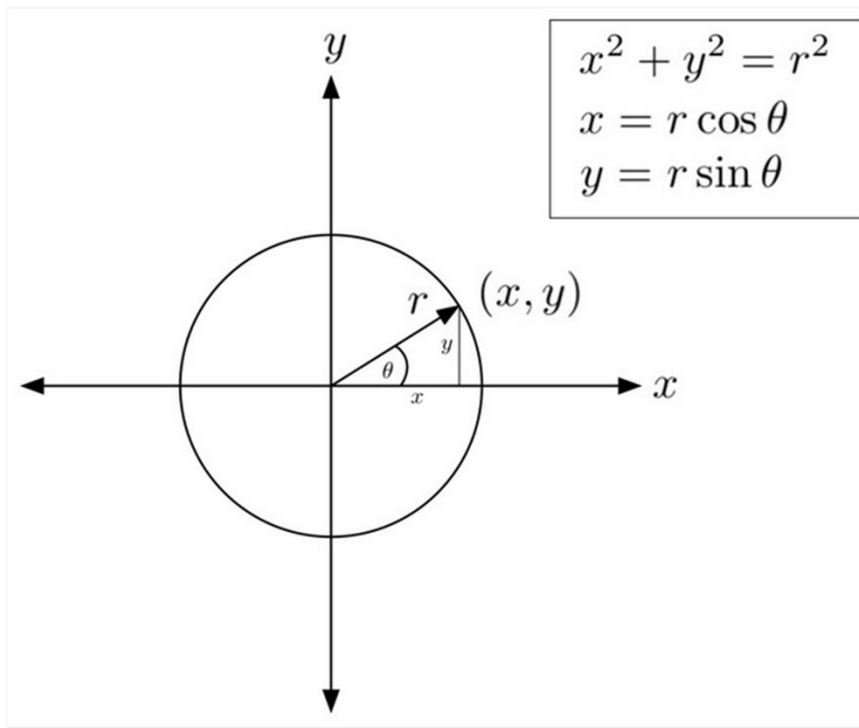
Rotation

Everything rotates



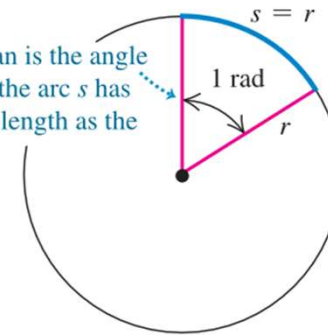
Rigid solids

Coordinates



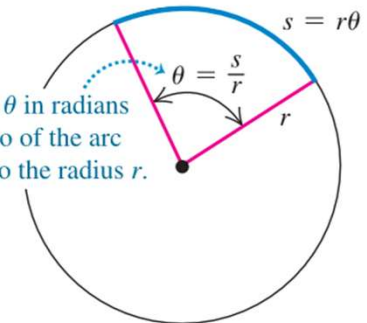
(a)

One radian is the angle at which the arc s has the same length as the radius r .



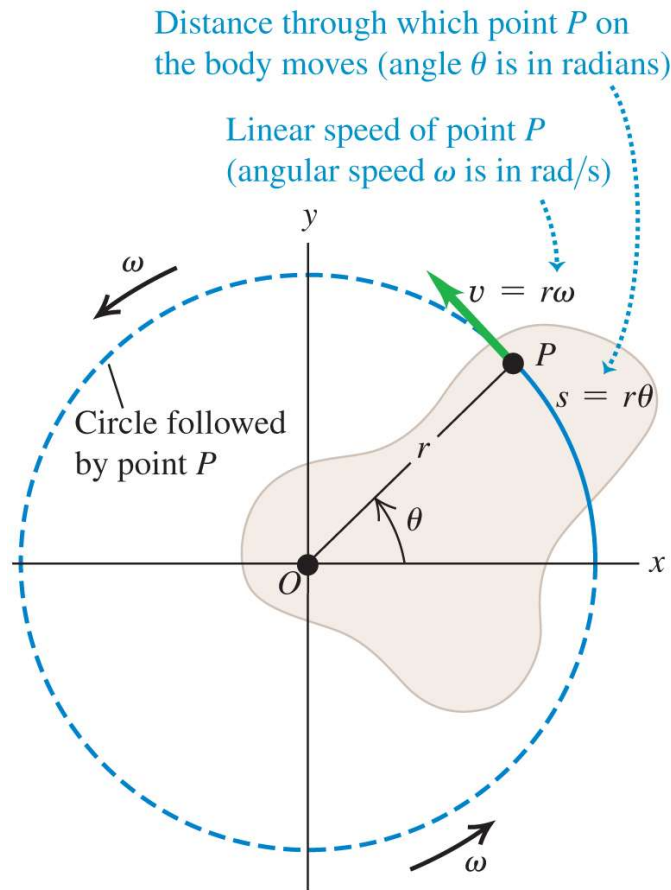
(b)

An angle θ in radians is the ratio of the arc length s to the radius r .



Speed for rotating bodies

$$v = \omega \cdot R$$



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$$\omega = \frac{2\pi \cdot \text{rad}}{T(s)}$$

T is the period for a complete rotation

Comparison of linear and rotary motion

TABLE 9.1 Comparison of Linear and Angular Motions with Constant Acceleration

Straight-Line Motion with Constant Linear Acceleration

$$a_x = \text{constant}$$

$$v_x = v_{0x} + a_x t$$

$$x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$$

$$x - x_0 = \frac{1}{2}(v_{0x} + v_x)t$$

Fixed-Axis Rotation with Constant Angular Acceleration

$$\alpha_z = \text{constant}$$

$$\omega_z = \omega_{0z} + \alpha_z t$$

$$\theta = \theta_0 + \omega_{0z} t + \frac{1}{2} \alpha_z t^2$$

$$\omega_z^2 = \omega_{0z}^2 + 2\alpha_z(\theta - \theta_0)$$

$$\theta - \theta_0 = \frac{1}{2}(\omega_{0z} + \omega_z)t$$

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Direction of rotation

You leave a building pushing through a door. The door is hinged on the right hand side.

In what direction does the angular velocity, $\vec{\omega}$, of the door point?

- ☐ It points to the right.
- ☐ It points to the left.
- ☐ It points up.
- ☐ It points down.



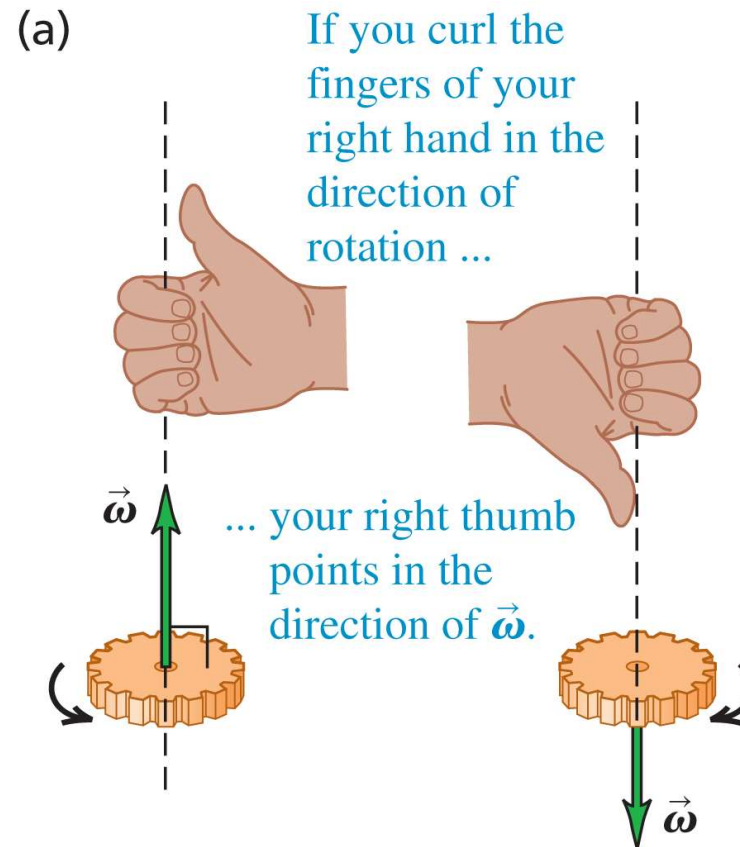
Direction of rotation

You are driving on your bicycle while you break.

In what direction does the angular velocity, $\vec{\omega}$, and the angular acceleration of the wheels, $\vec{\alpha}$, point?

- ☐ Both point to the left.
- ☐ Both point to the right.
- ☐ $\vec{\omega}$ point to the left and $\vec{\alpha}$ points to the right.
- ☐ $\vec{\omega}$ points to the right and $\vec{\alpha}$ points to the left.

Direction of rotation



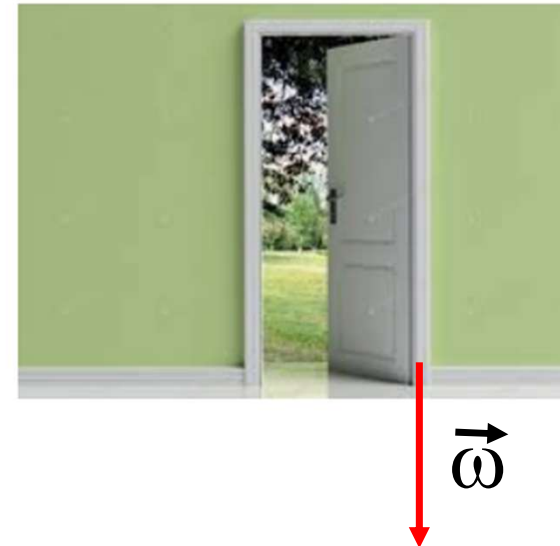
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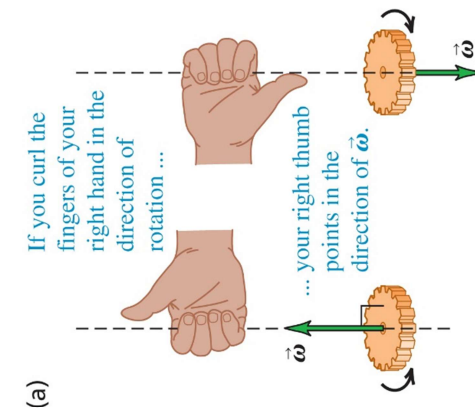


Direction of rotation

You are driving on your bicycle while you break.

In what direction does the angular velocity, $\vec{\omega}$, and the angular acceleration of the wheels, $\vec{\alpha}$, point?

- ☐ Both point to the left.
- ☐ Both point to the right.
- ☒ $\vec{\omega}$ point to the left and $\vec{\alpha}$ points to the right.
- ☐ $\vec{\omega}$ points to the right and $\vec{\alpha}$ points to the left.



Speed for rotating bodies

A wheel rolls to the right on a straight road. The center of mass has velocity $\vec{v}_0$.

☐

The point on the edge of the wheel above the center of mass has the velocity

☐

The point in contact with the road has velocity

☐

The point between the center of mass and the contact point with the road has the speed

☐

The left most point on the wheel has the speed

1. $\vec{v} = \vec{0}$

2. $\vec{v} = \vec{v}_0$

3. $\vec{v} = \frac{1}{2}\vec{v}_0$

4. $\vec{v} = 2\vec{v}_0$

5. $v < v_0$

6. $v > v_0$

7. $v = v_0$

Speed for rotating bodies

A wheel rolls to the right on a straight road. The center of mass has velocity $\vec{v}_0$.

☒

The point on the edge of the wheel above the center of mass has the velocity

☐

The point in contact with the road has velocity

☐

The point between the center of mass and the contact point with the road has the speed

☐

The left most point on the wheel has the speed

1. $\vec{v} = \vec{0}$

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3. $\vec{v} = \frac{1}{2}\vec{v}_0$

4. $\vec{v} = 2\vec{v}_0$

5. $v < v_0$

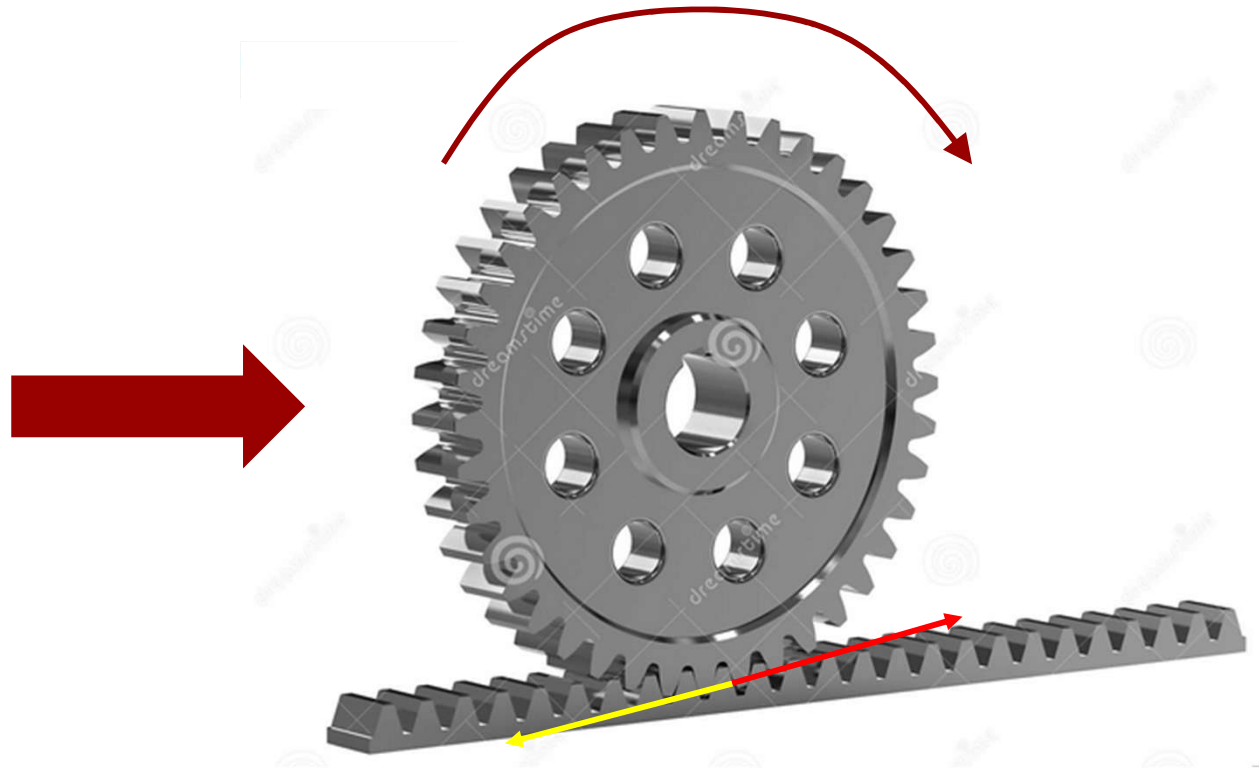
6. $v > v_0$

7. $v = v_0$

Velocity

Speed

Speed for rotating bodies



Rolling without slipping: The distance that the wheel advances on the road (translation) when the wheel completes a full rotation is equal to the length of the circumference of the wheel

Speed for rotating bodies



v_{CM} Translational speed of the center of mass
(which coincides with the axis of the wheel)

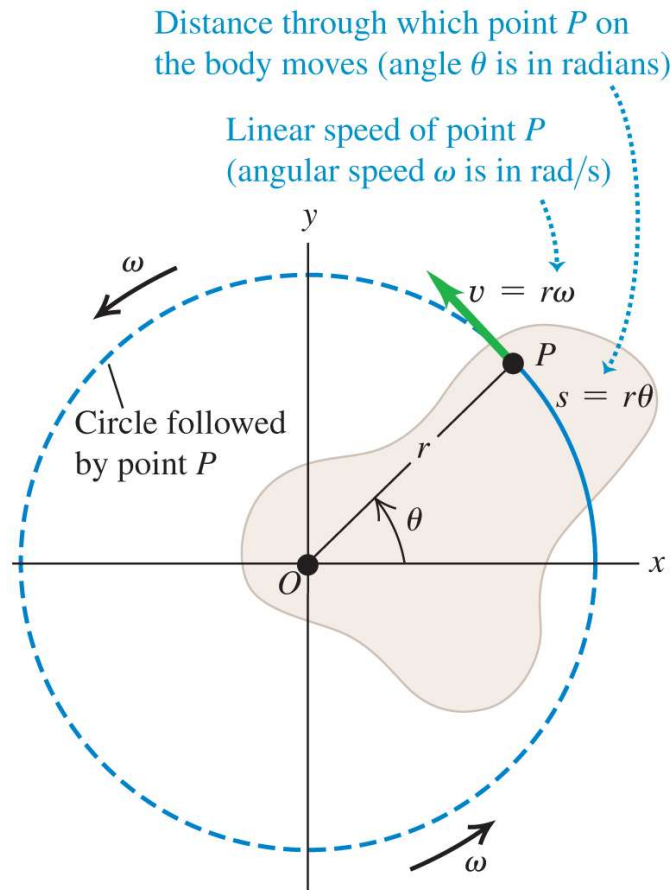
$$v_{CM} = \frac{\text{Length of the circumference}}{\text{Time for a full rotation (period)}}$$

$$v_{CM} = \frac{2\pi R}{T}$$

Rolling without slipping: The distance that the wheel advances on the road (translation) when the wheel completes a full rotation is equal to the length of the circumference of the wheel

Speed for rotating bodies

$$v_{LR} = \omega \cdot R$$



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$$\omega = \frac{2\pi \cdot \text{rad}}{T(s)}$$

T is the period for a complete rotation

Speed for rotating bodies



v_{CM} Translational speed of the center of mass
(which coincides with the axis of the wheel)

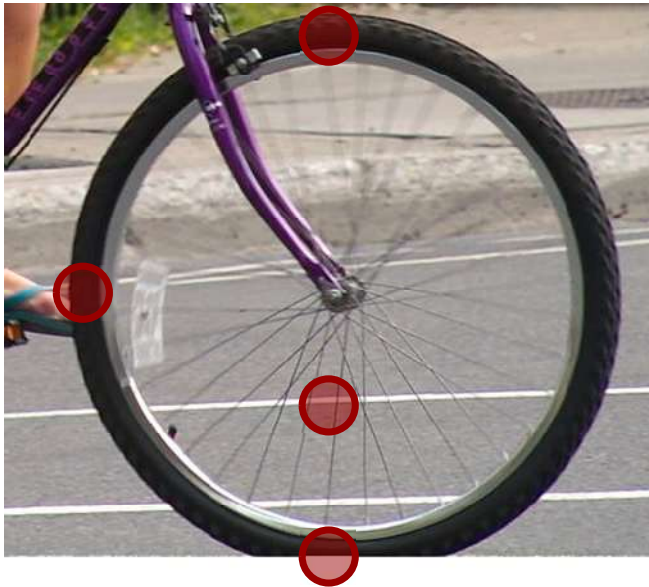
$$v_{CM} = \frac{\text{Length of the circumference}}{\text{Time for a full rotation (period)}}$$

$$v_{CM} = \frac{2\pi R}{T} = \frac{2\pi R}{T} = \omega \bullet R$$

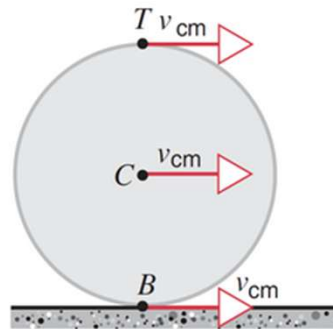
ω

Rolling without slipping: The distance that the wheel advances on the road (translation) when the wheel completes a full rotation is equal to the length of the circumference of the wheel

Speed for rotating bodies

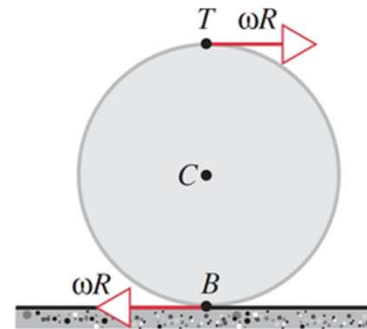


Translation



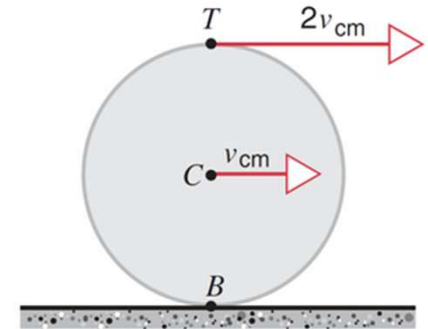
+

Rotation



=

Total Movement
of a point of the
wheel

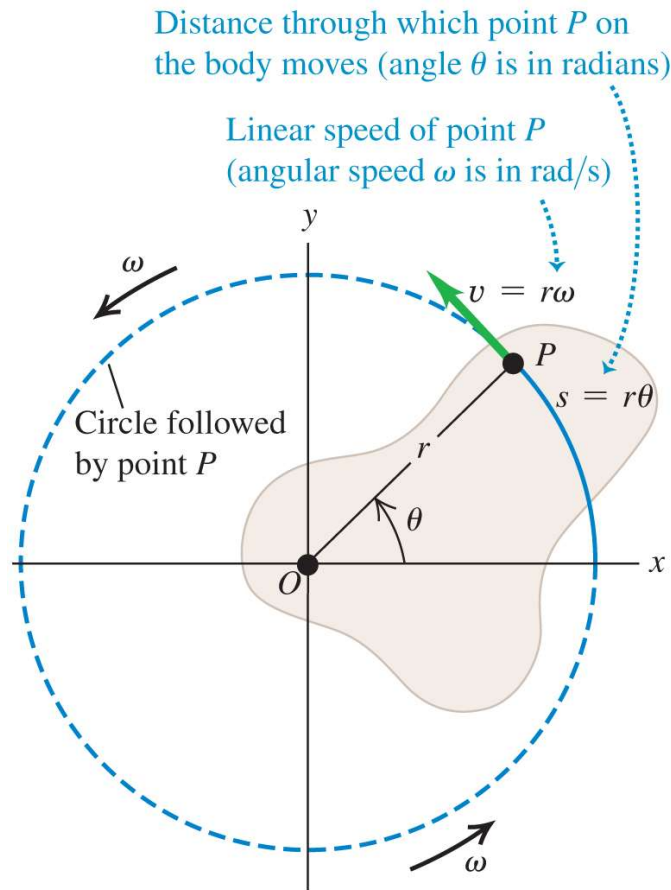


$$v_{CM} = \omega \cdot R$$

<https://courses.lumenlearning.com/suny-osuniversityphysics/chapter/11-1-rolling-motion/>

Speed for rotating bodies

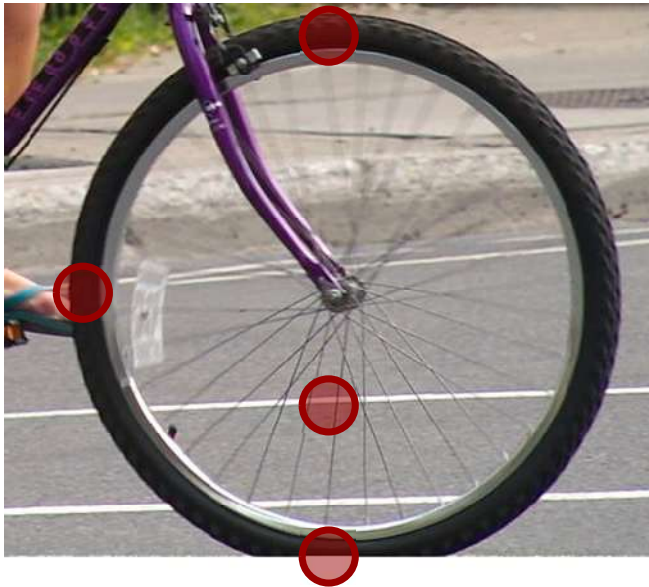
$$v_{LR} = \omega \cdot R$$



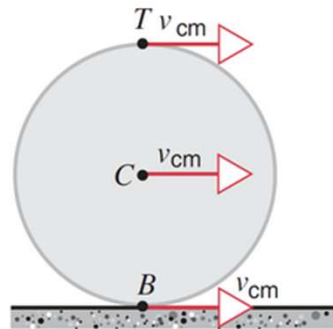
$$\omega = \frac{2\pi \cdot \text{rad}}{T(s)}$$

T is the period for a complete rotation

Speed for rotating bodies



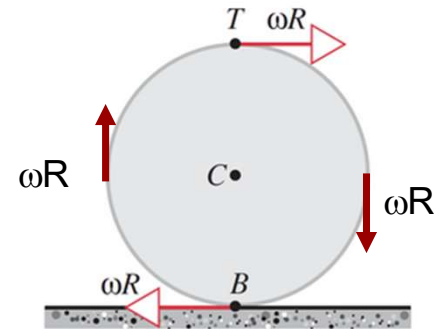
Translation



$$v_{CM} = \omega \cdot R$$

+

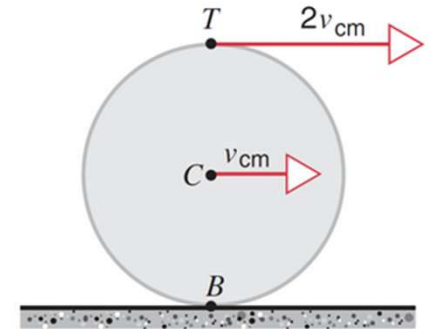
Rotation



$$v_{LR} = \omega \cdot R$$

=

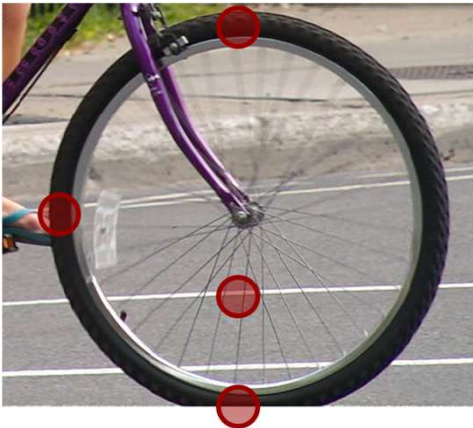
Total Movement
of a point of the
wheel



<https://courses.lumenlearning.com/suny-osuniversityphysics/chapter/11-1-rolling-motion/>

Speed for rotating bodies

A wheel rolls to the right on a straight road. The center of mass has velocity $\vec{v}_0$.



4 ▾

The point on the edge of the wheel above the center of mass has the velocity

1 ▾

The point in contact with the road has velocity

5 ▾

The point between the center of mass and the contact point with the road has the speed

6 ▾

The left most point on the wheel has the speed

1. $\vec{v} = \vec{0}$

2. $\vec{v} = \vec{v}_0$

3. $\vec{v} = \frac{1}{2}\vec{v}_0$

4. $\vec{v} = 2\vec{v}_0$

5. $v < v_0$

6. $v > v_0$

7. $v = v_0$

Acceleration in rotating bodies

Tangential acceleration of a point on a rotating rigid body

Distance of that point from rotation axis

$$a_{\text{tan}} = \frac{dv}{dt} = r \frac{d\omega}{dt} = r\alpha$$

Rate of change of linear speed of that point

Rate of change of angular speed of body

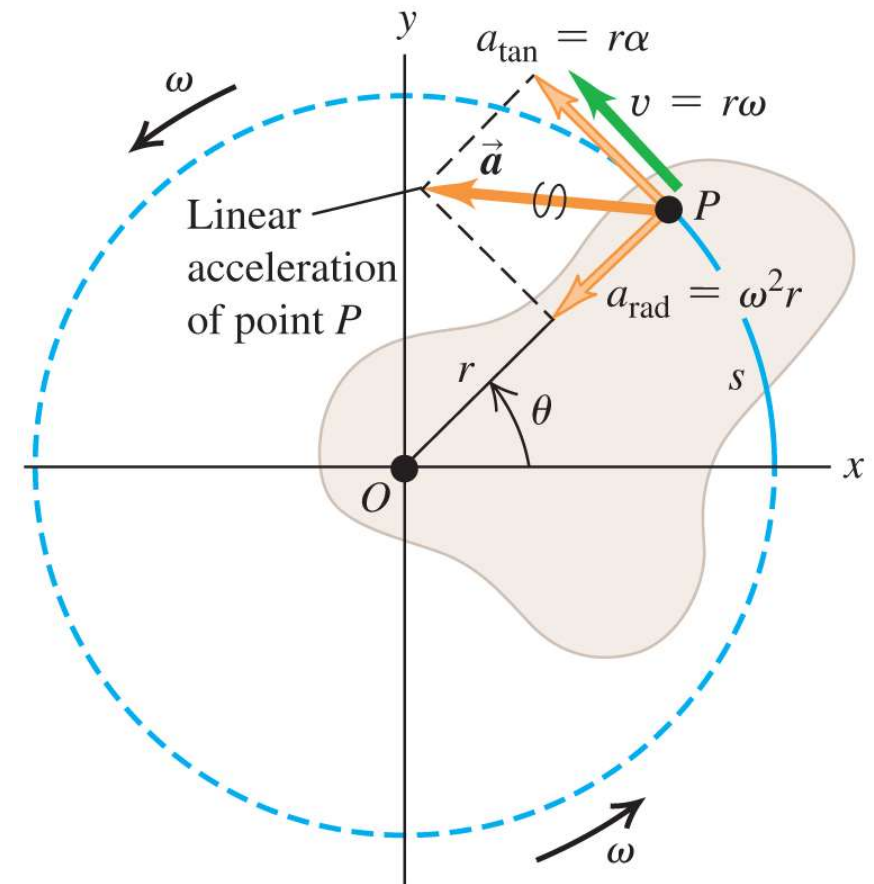
Centripetal acceleration of a point on a rotating rigid body

Linear speed of that point

Angular speed of body

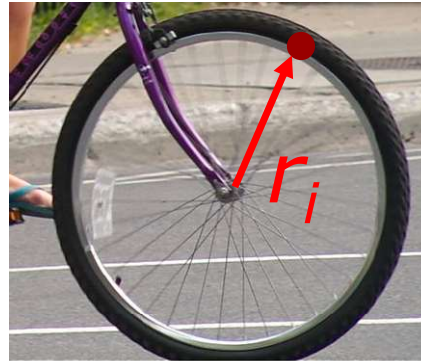
$$a_{\text{rad}} = \frac{v^2}{r} = \omega^2 r$$

Distance of that point from rotation axis

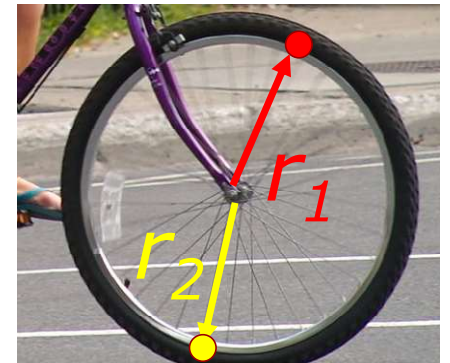


Energy in rotation

$$\frac{1}{2}m_i v_i^2 = \frac{1}{2}m_i r_i^2 \omega^2$$



$$K = \frac{1}{2}m_1 r_1^2 \omega^2 + \frac{1}{2}m_2 r_2^2 \omega^2 + \dots = \sum_i \frac{1}{2}m_i r_i^2 \omega^2$$



Energy in rotation

$$K = \frac{1}{2}m_1r_1^2\omega^2 + \frac{1}{2}m_2r_2^2\omega^2 + \dots = \sum_i \frac{1}{2}m_ir_i^2\omega^2$$

Moment of inertia

of a body for a given rotation axis

Masses of the particles that make up the body

$$I = m_1r_1^2 + m_2r_2^2 + \dots = \sum_i m_ir_i^2$$

Perpendicular distances of the particles from rotation axis

Rotational kinetic energy

of a rigid body rotating around an axis

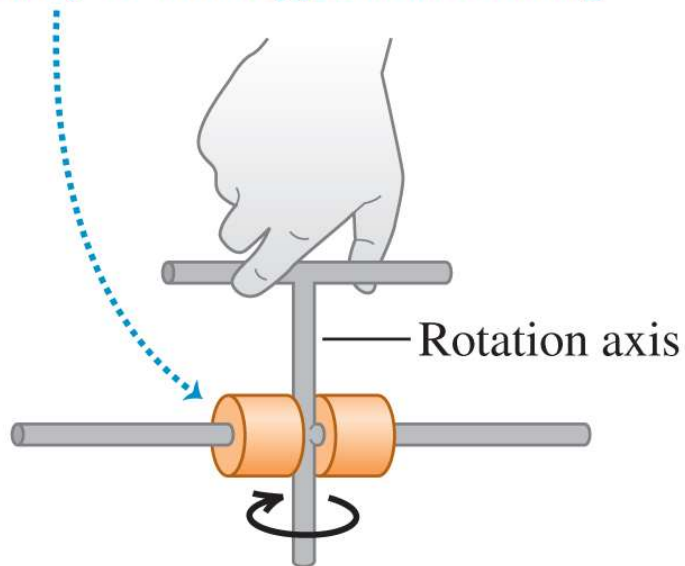
$$K = \frac{1}{2}I\omega^2$$

Angular speed of body

Moment of inertia of body for given rotation axis

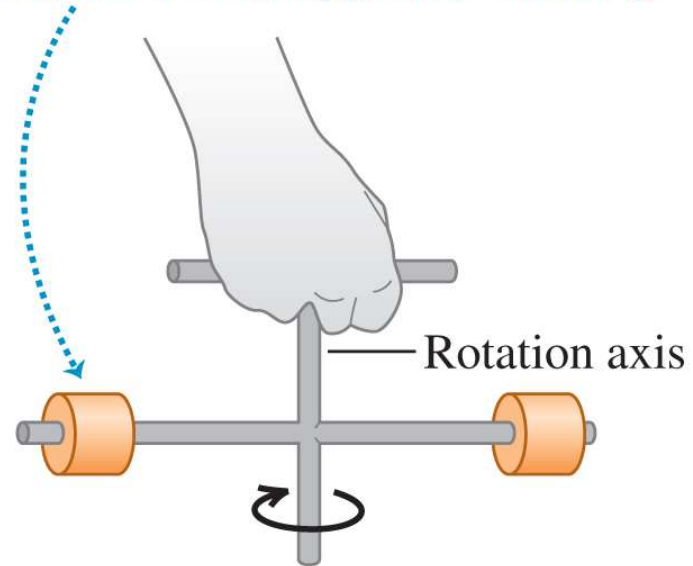
Illustration of moment of inertia

- Mass close to axis
- Small moment of inertia
- Easy to start apparatus rotating



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- Mass farther from axis
- Greater moment of inertia
- Harder to start apparatus rotating



Moment of inertia

Moment of inertia of light rod with heavy spheres at the ends.

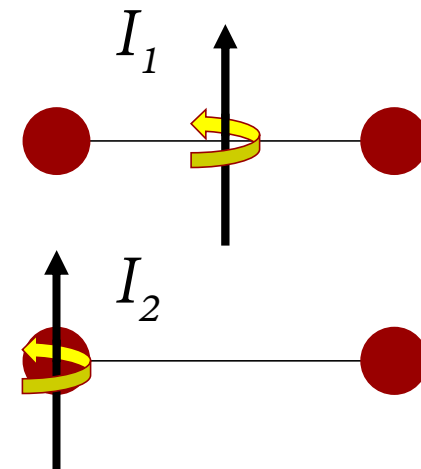
A light rod (the mass of the rod can be ignored) have to identical heavy solid spheres fixed at the ends. The moment of inertia with respect to an axis perpendicular to the direction of the rod passing through the center is called I_1 . The moment of inertia of the system with respect to an axis perpendicular to the direction of the rod passing through one end is called I_2 .

What is the relation between the two moments of inertia?

☐ $I_1 = I_2$

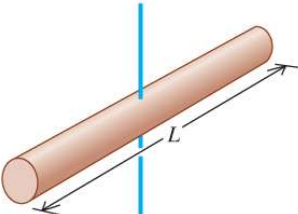
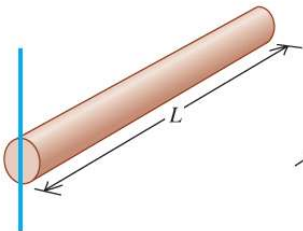
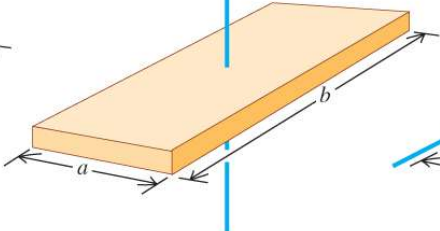
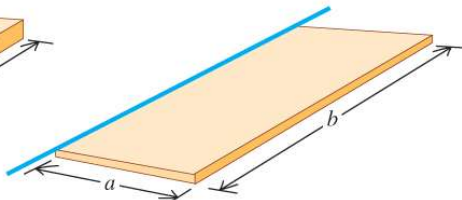
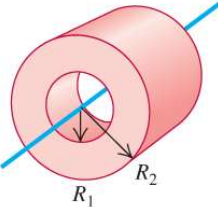
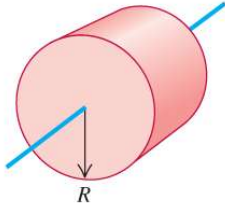
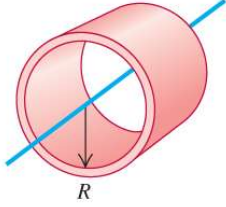
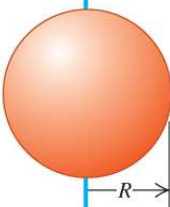
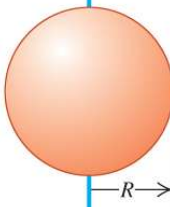
☐ $I_1 > I_2$

☐ $I_1 < I_2$



Moment of inertia

TABLE 9.2 Moments of Inertia of Various Bodies

(a) Slender rod, axis through center $I = \frac{1}{12}ML^2$ 	(b) Slender rod, axis through one end $I = \frac{1}{3}ML^2$ 	(c) Rectangular plate, axis through center $I = \frac{1}{12}M(a^2 + b^2)$ 	(d) Thin rectangular plate, axis along edge $I = \frac{1}{3}Ma^2$ 	
(e) Hollow cylinder $I = \frac{1}{2}M(R_1^2 + R_2^2)$ 	(f) Solid cylinder $I = \frac{1}{2}MR^2$ 	(g) Thin-walled hollow cylinder $I = MR^2$ 	(h) Solid sphere $I = \frac{2}{5}MR^2$ 	(i) Thin-walled hollow sphere $I = \frac{2}{3}MR^2$ 

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Moment of inertia

A wheel that is a solid, homogeneous disk with mass m and radius r , rolls with constant velocity on a horizontal surface. The speed is v .

The disk's translational kinetic energy is K_{trans} and its rotational kinetic energy is K_{rot} .

What is the relation between the two contributions to the total kinetic energy?

- ☐ $K_{\text{trans}} = K_{\text{rot}}$
- ☐ $K_{\text{trans}} < K_{\text{rot}}$
- ☐ $K_{\text{trans}} > K_{\text{rot}}$

Moment of inertia

A wheel that is a solid, homogeneous disk with mass m and radius r , rolls with constant velocity on a horizontal surface. The speed is v .

The disk's translational kinetic energy is K_{trans} and its rotational kinetic energy is K_{rot} .

What is the relation between the two contributions to the total kinetic energy?

☐ $K_{\text{trans}} = K_{\text{rot}}$

☐ $K_{\text{trans}} < K_{\text{rot}}$

☐ $K_{\text{trans}} > K_{\text{rot}}$

$$K_{\text{trans}} = \frac{1}{2} M v_{\text{CM}}^2$$

Speed for rotating bodies



v_{CM} Translational speed of the center of mass
(which coincides with the axis of the wheel)

$$v_{CM} = \frac{\text{Length of the circumference}}{\text{Time for a full rotation (period)}}$$

$$v_{CM} = \frac{2\pi R}{T} = \frac{2\pi R}{T} = \omega \bullet R$$

ω

Rolling without slipping: The distance that the wheel advances on the road (translation) when the wheel completes a full rotation is equal to the length of the circumference of the wheel

Moment of inertia

A wheel that is a solid, homogeneous disk with mass m and radius r , rolls with constant velocity on a horizontal surface. The speed is v .

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☐ $K_{\text{trans}} = K_{\text{rot}}$

☐ $K_{\text{trans}} < K_{\text{rot}}$

☐ $K_{\text{trans}} > K_{\text{rot}}$

$$K_{\text{trans}} = \frac{1}{2} M v_{\text{CM}}^2$$

$$v_{\text{CM}} = \omega \cdot R$$

Energy in rotation

$$K = \frac{1}{2}m_1r_1^2\omega^2 + \frac{1}{2}m_2r_2^2\omega^2 + \dots = \sum_i \frac{1}{2}m_ir_i^2\omega^2$$

Moment of inertia

of a body for a given rotation axis

Masses of the particles that make up the body

$$I = m_1r_1^2 + m_2r_2^2 + \dots = \sum_i m_ir_i^2$$

Perpendicular distances of the particles from rotation axis

Rotational kinetic energy

of a rigid body rotating around an axis

$$K = \frac{1}{2}I\omega^2$$

Angular speed of body

Moment of inertia

of body for given rotation axis

Moment of inertia

A wheel that is a solid, homogeneous disk with mass m and radius r , rolls with constant velocity on a horizontal surface. The speed is v .

The disk's translational kinetic energy is K_{trans} and its rotational kinetic energy is K_{rot} .

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☐ $K_{\text{trans}} = K_{\text{rot}}$

☐ $K_{\text{trans}} < K_{\text{rot}}$

☐ $K_{\text{trans}} > K_{\text{rot}}$

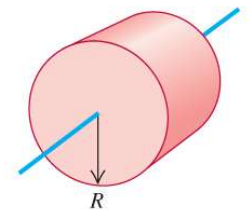
$$K_{\text{trans}} = \frac{1}{2} M v_{\text{CM}}^2$$

$$v_{\text{CM}} = \omega \cdot R$$

$$K_{\text{rot}} = \frac{1}{2} I \omega^2$$

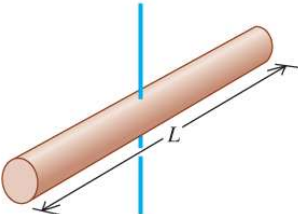
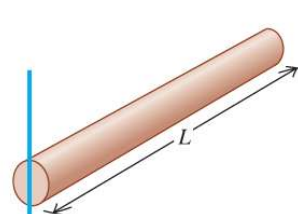
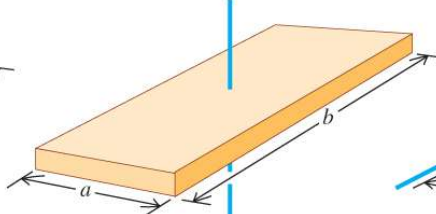
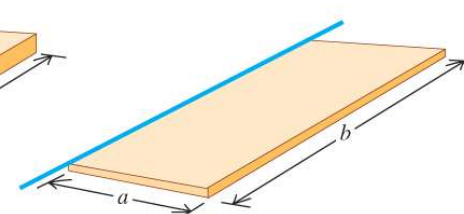
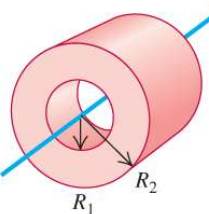
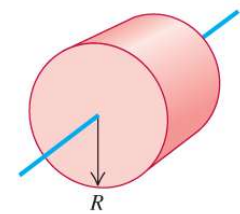
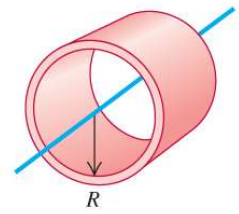
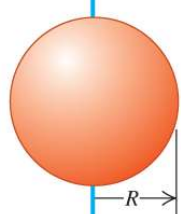
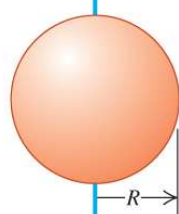
(f) Solid cylinder

$$I = \frac{1}{2} M R^2$$



Moment of inertia

TABLE 9.2 Moments of Inertia of Various Bodies

(a) Slender rod, axis through center $I = \frac{1}{12}ML^2$ 	(b) Slender rod, axis through one end $I = \frac{1}{3}ML^2$ 	(c) Rectangular plate, axis through center $I = \frac{1}{12}M(a^2 + b^2)$ 	(d) Thin rectangular plate, axis along edge $I = \frac{1}{3}Ma^2$ 	
(e) Hollow cylinder $I = \frac{1}{2}M(R_1^2 + R_2^2)$ 	(f) Solid cylinder $I = \frac{1}{2}MR^2$ 	(g) Thin-walled hollow cylinder $I = MR^2$ 	(h) Solid sphere $I = \frac{2}{5}MR^2$ 	(i) Thin-walled hollow sphere $I = \frac{2}{3}MR^2$ 

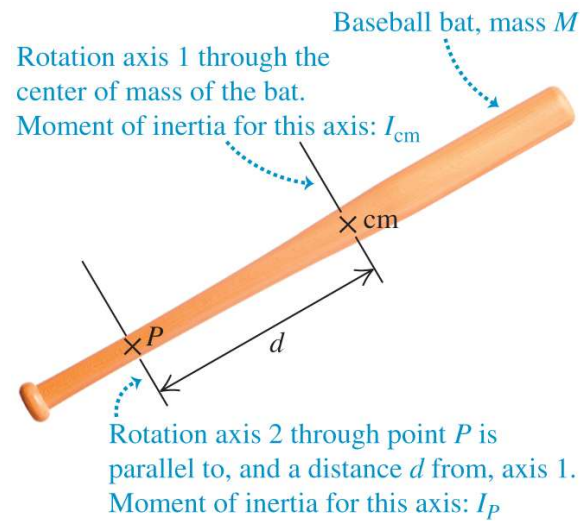
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Exercises

A rod with length L and mass M can rotate freely about an axis through one of its ends ($x = 0$). The rod is not homogeneous, and has a density $\rho(x) = \alpha \cdot x$, where the density is mass per length.

- a) What is the SI unit of the parameter α ?
- b) Determine the rod's moment of inertia about the rotation axis.

Parallel-axis (Steiner) theorem



Parallel-axis theorem: $I_P = I_{cm} + Md^2$

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Parallel-axis theorem:

Moment of inertia of a body for a rotation axis through point P

$$I_P = I_{cm} + Md^2$$

Moment of inertia of body for a parallel axis through center of mass

Mass of body

Distance between two parallel axes

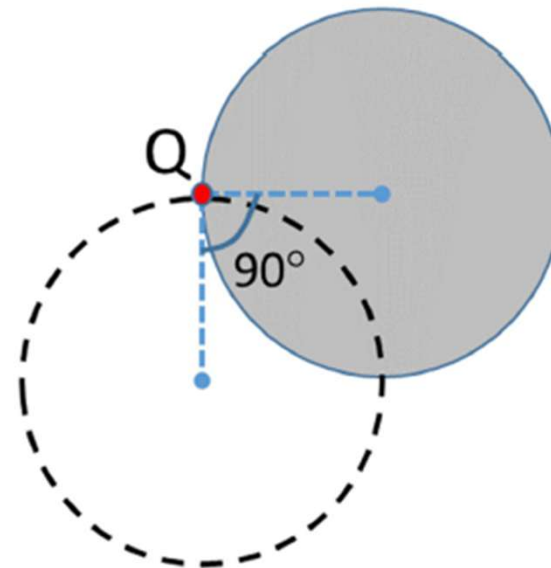
(9.19)

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Example: Exam May 2022

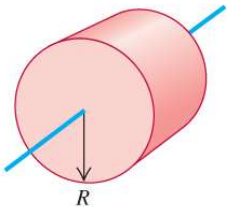
A uniform solid disk with radius $R = 30$ cm and mass $M = 1$ kg. can freely rotate through an axis Q that passes through its edge (see Figure). Initially the disk is kept at rest in the position shown by the grey circle in the figure. Then the disk is freed and starts moving.

Data: Moment of inertia of a disk through an axis passing through its center of mass $I_{CM} = \frac{1}{2} M R^2$. Use $g = 10$ m/s².



(f) Solid cylinder

$$I = \frac{1}{2} M R^2$$



What is the speed of the center of mass of the disk when it reaches the position indicated by the dashed line circle?